Brian Kim

College Park, MD | kimbrian@umd.edu | kimbrianj.github.io | kimbrianj

Education

PhD **University of California, Los Angeles**, Statistics — Los Angeles, CA, Sept 2012 – Oct 2017
- Dissertation: Population Size Estimation using Multiple Respondent-Driven Samples

BA **Amherst College**, Mathematics & Philosophy — Amherst, MA, Sept 2008 – June 2012
- Graduated with Honors in Mathematics
- Thesis: Using Mode Identification Clustering Methods with Pitch F/X Data

Research Interests

- Data Science Education; Social Network Analysis; Network Sampling Methods; Respondent-Driven Sampling; Population Size Estimation: Capture-Recapture and Multiple List methods; Social Science Applications of Machine Learning

Experience

College of Behavioral and Social Sciences (BSOS), University of Maryland, Co-Director, Social Data Science Major — College Park, MD, Oct 2021 – present, 4 years 6 months
- Led the development of a new major in Social Data Science (officially launched Fall 2022), including working across departments to build an appropriate curriculum, managing the budget, and setting up infrastructure.
- Developed new courses to be part of Social Data Science, including SDSB233: Data Science for Social Sciences and SDSB326: Python Programming for Social Science.
- Launched a new undergraduate Data Curation Fellowship experience in which students identify, obtain, clean, and curate datasets to be used within social science courses, with a public-facing data resource webpage ([https://bsos-data.umd.edu]).
- Coordinated across colleges with the College of Information Co-Director to set up advising, teaching workload, and major requirements for students.

Joint Program in Survey Methodology (JPSM), University of Maryland, Director of Graduate Studies — College Park, MD, Sept 2024 – present, 1 year 7 months
- Oversaw updates to the Master's in Survey and Data Science program, including adding SURV613: Machine Learning for Social Science as a required course and incorporating neural networks and LLMs into course material.
- Assisted graduate students in finding funding through grants or teaching assistant opportunities on campus.
- Chaired the Master's Admissions committee and coordinated with the Director of JPSM on recruitment efforts.

Social Data Science; Joint Program in Survey Methodology, Associate Research Professor — College Park, MD; July 2024 – present; 1 year 9 months

- Developed and implemented innovative teaching assignments and tools including: cloud-based programming workbooks using JupyterHub (multiple classes); a board game decision analysis project for an undergraduate class (INST 354: Decision Making for Information Science); a data disclosure-proofing project (SURV 622: Fundamentals of Data Collection); and a custom R package containing tutorials (SURV 613: Machine Learning for Social Science).
- Taught classes on introductory statistics, data science, machine learning, and programming for both undergraduate and graduate students, utilizing traditional and flipped classroom formats in online, in-person, and HyFlex modes.
- As part of an NSF grant, managed and oversaw the development and day-to-day operations of data analytics training courses for working professionals and graduate students at the National Center for Science and Engineering Statistics (NCSES) at the National Science Foundation (NSF).

Social Data Science; Joint Program in Survey Methodology, Assistant Research Professor — College Park, MD; Aug 2020 – July 2024; 4 years

Joint Program in Survey Methodology, University of Maryland, Lecturer — College Park, MD; Oct 2017 – Aug 2020; 2 years 11 months

The Coleridge Initiative, Consultant/Facilitator — Sept 2020 – Jan 2022; 1 year 5 months

- Provided guidance as a facilitator with the Applied Data Analytics program for teaching record linkage, text analysis, network analysis, and machine learning to working professionals in public policy.
- Consulted on a research projects using machine learning for imputation with large scale transaction-level purchasing data.
- Developed modularized lessons and notebooks for introductory statistics and programming in Python, R and SQL.

UCLA Institute for Digital Research and Education, Statistical Consultant — Los Angeles, CA; May 2014 – June 2016; 2 years 2 months

- Assisted graduate students and professors in Psychology, Economics, Education, and other fields in cleaning, merging, and preparing data for analysis.
- Helped clients with their dissertations, publications, and other projects using methods such as mixed modeling, principal components analysis, simulation, and more using a variety of statistical software.

UCLA Institute for Pure and Applied Mathematics, Academic Mentor — UCLA; 2015 – 2017; 2 years

- Mentored a team of four undergraduate students with a research project for the Research in Industrial Projects for Students (RIPS) summer program in which an industry sponsor provides a real problem of interest.
- Provided a cooperative team environment for research by encouraging collaboration between students and advising students throughout the full research lifecycle.

Publications

Using a Stopping Rule to Optimize Cost-Quality Tradeoffs in a Large, Mixed-Mode Survey: A Simulation Study
J. Wagner, B. West, **B. Kim**, D. Suolang, C. Engstrom, J. Sinibaldi
www.doi.org/10.1177/0282423X241287452

Hidden Population Size Estimation and Diagnostics Using Two Respondent-Driven Samples with Applications in Armenia
B. Kim, L. G. Johnston, T. Grigoryan, A. Papoyan, S. Grigoryan, K. R. McLaughlin
www.doi.org/10.1002/bimj.202200136

Using Geographical Data and Rolling Statistics for Diagnostics of Respondent-Driven Sampling
B. Kim, M. Ogwal, E. Sande, H. Kiyingi, D. Serwadda, W. Hladik
www.doi.org/10.1016/j.socnet.2020.05.001

Population Size Estimation Using Multiple Respondent-Driven Sampling Surveys
B. Kim, M. Handcock
www.doi.org/10.1093/jssam/smz055

Easy-to-Use Cloud Computing for Teaching Data Science
B. Kim, G. Henke
www.doi.org/10.1080/10691898.2020.1860726

Partnering with a global platform to inform research and public policy making
F. Kreuter, N. Barkay, A. Bilinski, A. Bradford, S. Chiu, R. Eliat, J. Fan, T. Galili, D. Haimovich, **B. Kim**, S. LaRocca, Y. Li, K. Morris, S. Presser, T. Sarig, J. A. Salomon, K. Stewart, E. A. Stuart, R. Tibshirani
www.doi.org/10.18148/srm/2020.v14i2.7761

Software and Data

sspse: Estimating Hidden Population Size using Respondent Driven Sampling Data
M. Handcock, K. Gile, **B. Kim**, K. McLaughlin
cran.r-project.org/package=sspse

The University of Maryland social data science center global COVID-19 trends and impact survey, in partnership with Facebook
J. Fan, Y. Li, K. Stewart, A. R. Kommareddy, A. Garcia, J. O'Brien, A. Bradford, X. Deng, S. Chiu, F. Kreuter, N. Barkay, A. Bilinski, **B. Kim**, T. Galili, D. Haimovich, S. LaRocca, S. Presser, K. Morris, J. Salomon, D. Vannette
covidmap.umd.edu/api.html

Book Chapters and Non-Refereed Publications

The Value of Science: Special Theme. Issue 4.2
J. Lane, **B. Kim**, F. Kreuter, A. Nunez
10.1162/99608f92.4a01213d

Scaling Up Data Science for the Social Sciences
B. Kim
10.1162/99608f92.d3f14ea4

Workbooks
B. Kim, C. Kern, J. Morgan, C. Hunter, A. Kumar
textbook.coleridgeinitiative.org/chap-workbooks.html

Selected Conference Presentations

A Hybrid Household and Respondent-Driven Sampling Strategy for Reaching Underrepresented Groups
B. Kim, A. Gard

Machine Learning Model Selection for Complex Sample Survey Data
B. Kim

Data 8 Adoption at University of Maryland
B. Kim

Population Size Estimation Using Multiple Respondent-Driven Sampling Surveys
B. Kim, M. Handcock

Teaching Data Science Using Jupyter Notebooks and Binder
B. Kim

Assessing Respondent-Driven Sampling Using Geographical Data
B. Kim, M. Ogwal, E. Sande, H. Kiyingi, D. Serwadda, W. Hladik

Grants and Awards

Advancing Computational Expertise in Neuroscience and Psychology (co-PI) May 2025 – May 2026
UMD Artificial Intelligence Interdisciplinary Institute: Course Development Grant
- Funding: $10,000

Cloud-based Active Learning for BSOS Statistics and Data Science Courses (PI) Sept 2022 – Aug 2023
UMD Teaching and Learning Transformation Center: Experiential Learning Grant
- Funding: $135,573.72

An Evaluation of Sampling Methodologies for Reaching Underrepresented Groups (PI) June 2022 – May 2023
University of Maryland BSOS Dean's Research Initiative
- Funding: $19,999

Global Privacy Monitor (GPM): Learning from External Privacy Stakeholders June 2021 – May 2024
Facebook, Inc.
- Funding: $2,994,731

Modernizing NCSES Data Collection Approaches (co-PI) Sept 2020 – Aug 2021
National Science Foundation, NCSES BAA
- Funding: $88,802

Collaborative Research: Impacts of Hard/Soft Skills on STEM Workforce Trajectories (co-PI) May 2020 – May 2025
National Science Foundation, ECR: 1956114
- Funding: $401,716

Linked Dataset and Virtual Research Data Center (Co-PI) Aug 2019 – Feb 2024
National Agricultural Statistics Service
- Funding: $653,490

Teaching Innovations Grant June 2020 – Aug 2020
University of Maryland
- Funding: $18,000

Teaching Innovations Grant June 2020 – Aug 2020
University of Maryland
- Funding: $5,000

Developing a New Course in Data Science: Introduction to Data Science for Social Sciences (PI) July 2019 – May 2020
UMD Year of Data Science
- Funding: $17,500

Feasibility Testing of a Respondent-Driven Sampling Approach to Recruit Human Trafficking Victims Dec 2018 – July 2019
Governor's Office of Crime Control and Prevention
- Funding: $30,000

Dissertation Year Fellowship 2016 – 2016
University of California, Los Angeles

Collegium of University Teaching Fellows 2016 – 2016
University of California, Los Angeles

Graduate Dean's Scholar Award 2012 – 2014
University of California, Los Angeles

Other Academic Work

Guest Editor 2022 – 2022
- Harvard Data Science Review, Special Theme: Value of Science. Issue 4.2.

Reviewer 2018 – present
- Journal of Statistics and Data Science Education
- Journal of Survey Statistics and Methodology
- Journal of Statistical Software
- Public Opinion Quarterly
- Harvard Data Science Review
- Computational Statistics and Data Analysis
- Journal of Official Statistics

Teaching

University of Maryland, Instructor Oct 2017 – present
- SDSB180: **Introduction to Data Management** (Fall 2024, Spring 2025, Fall 2025, Spring 2026)
- SDSB123: **Social Data Science: Pathways and Applications** (Fall 2024, Spring 2025, Fall 2025, Spring 2026)
- SDSB326: **Python Programming for Social Science** (Spring 2024)
- SDSB233: **Data Science for Social Sciences** (Fall 2019, Spring 2020, Spring 2022, Fall 2022, Spring 2023)
- INST 314: **Statistics for Information Science** (Fall 2018)
- INST 354: **Decision Making for Information Science** (Spring 2019)
- SURV 613: **Machine Learning for Social Sciences** (Fall 2020, Fall 2021, Spring 2023, Spring 2024, Spring 2025, Spring 2026)
- SURV 622: **Fundamentals of Data Collection** (Spring 2020, Spring 2021, Spring 2022)
- SURV 673: **Introduction to Python & SQL** (Fall 2018, Spring 2019, Summer 2019, Fall 2019, Fall 2020, Summer 2020, Summer 2021)
- SURV 699M: **Review of Statistical Concepts** (Summer 2018, Summer 2019, Summer 2020, Summer 2021)
- SURV 701: **Analysis of Complex Sample Survey Data** (Spring 2021)
- SURV 727: **Fundamentals of Computing and Data Display** (Fall 2022, Fall 2023, Fall 2024, Fall 2025)
- SURV 736: **Introduction to Web Scraping with R** (Summer 2025)

University of California, Los Angeles, Instructor — Jan 2017 – Mar 2017

- Stat 98T: **Six Degrees of Separation: Studying the World Through Social Networks** (Winter 2017)

University of California, Los Angeles, Teaching Assistant — Sept 2013 – June 2016

- Stat 10: **Introduction to Statistical Reasoning** (Fall 2013, Winter 2014, Spring 2014, Fall 2014, Winter 2015, Spring 2015, Fall 2015, Winter 2016, Spring 2016)

Mentorship

Doctoral Students

Advisor

- Harold Gomes (in progress)

Doctoral Committee

Committee Member

- Yuting Chen, University of Maryland (2025)
- Ai Rene Ong, University of Michigan (2022)
- Jordan Epistola, University of Maryland (2022)

Graduate Students on Funded Projects

Supervisor

- Namit Shrivastava (January 2026 – May 2026)
- Ujjayini Das (September 2022 – August 2023)
- Joseph Hoskisson (September 2022 – August 2023)
- Franklin Yang (September 2021 – August 2022)
- Sofi Sinozich (September 2020 – August 2021)

Undergraduate Capstone Projects

Faculty Mentor"

- Pragya Kumar, Orla Collins, Giulia Hoorens van Heyningen: Assessing the Viability of Weighting Methods for Hybrid Sampling Design (2026)
- Danny Pham, Tariq Witherspoon, Andrew Shin: Machine Learning Model Selection with Complex Sample Survey Data (2026)

Service

- Faculty Coordinator, Joint Program in Survey Methodology Short Course Program
- Co-Chair, Social Data Science Program Committee
- Co-Chair, Department of Psychology Lecturer Search Committee
- Chair, Social Data Science Curriculum Committee
- Chair, Search Committee for Executive Director of the Center for Advances in Data and Measurement
- Member/Chair, JPSM/MPSM Masters Admissions Committee
- Member, Department of Economics Lecturer Search Committee
- Member, Department of Criminology and Criminal Justice Lecturer Search Committee
- Member, College of Information Studies Lecturer Search Committee
- Member, Social Data Science Advisor Search Committee
- Member, Social Data Science Steering Committee
- Member, JPSM/MPSM Teaching Assignment Committee

Skills

Statistical Software
Highly skilled in R, SQL and Python, including various packages such as Rcpp, caret, tidyverse, tidymodels, and more (in R); and pandas, scikit-learn, nltk, tensorflow, keras, and more (in Python).

Statistics
Highly skilled in a variety of methods, including but not limited to hypothesis testing, regression, mixed models, clustering analysis, social network analysis, social network models, network sampling methods, Monte Carlo simulation, Bayesian models, Machine Learning, text analysis, web scraping, and more.

Other Software
Skilled or proficient in the use of many other programs, including, but not limited to, Microsoft Word, Excel and Powerpoint; LaTeX.